



2016 HSC Course Assessment Task 3
(Trial Examination) June 21, 2016

- Commence each new question on a new page.
- Show all necessary working in every question. Marks may be deducted for illegible or incomplete working.

[illegible]

Section 1: Multiple Choice– 1 mark each.

Q1. The exact value of $\operatorname{cosec} \frac{7\pi}{6}$ is

(A) -2

(B) $-\frac{2}{\sqrt{3}}$

(C) $\frac{2}{\sqrt{3}}$

(D) 2

Q2. The value of

$$\sum_{i=10}^{21} 2i - 30$$

is

(A) 8

(B) 10

(C) 11

(D) 12

Q3. Which line is perpendicular to the line $3x + 4y + 7 = 0$?

(A) $4x + 3y - 7 = 0$

(B) $3x - 4y + 7 = 0$

(C) $8x - 6y - 7 = 0$

(D) $4x - 7y + 7 = 0$

Question 4 on the next page

Q4. Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = x \log_e x$ between $x = 1$ and $x = 3$

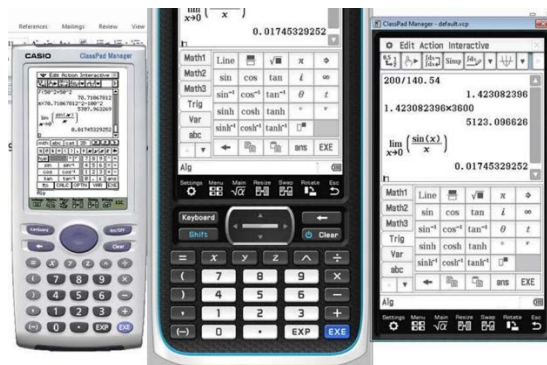
- (A) $\frac{1}{4}(\log_e 1 + 6 \log_e 1.5 + 4 \log_e 2 + 10 \log_e 2.5 + 3 \log_e 3)$
- (B) $\frac{1}{4}(\log_e 1 + 3 \log_e 1.5 + 4 \log_e 2 + 5 \log_e 2.5 + 3 \log_e 3)$
- (C) $\frac{1}{2}(\log_e 1 + 3 \log_e 1.5 + 4 \log_e 2 + 5 \log_e 2.5 + 3 \log_e 3)$
- (D) $\frac{1}{2}(\log_e 1 + 6 \log_e 1.5 + 4 \log_e 2 + 10 \log_e 2.5 + 3 \log_e 3)$

Q5. A student (not in NSW) is using technology in their exam to calculate a limit.

Their calculator tells them that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 0.017453 \dots$$

What happened?



- (A) The calculator made a rounding error.
- (B) The calculator is in degrees.
- (C) The calculator is in radians.
- (D) The calculator is in gradians.

Question 6 on the next page

Q6. A population of sea monkeys is observed to fluctuate according to the equation

$$\frac{dP}{dt} = 40 \sin(0.1t),$$

where P is the sea monkey population and t is the time in days.

During which day does the population first start to decrease?

(A) Day 15

(B) Day 16

(C) Day 31

(D) Day 32

Q7. The solution to $3x^2 + 7x > 6$ is

(A) $-\frac{1}{2} < x < -\frac{1}{3}$

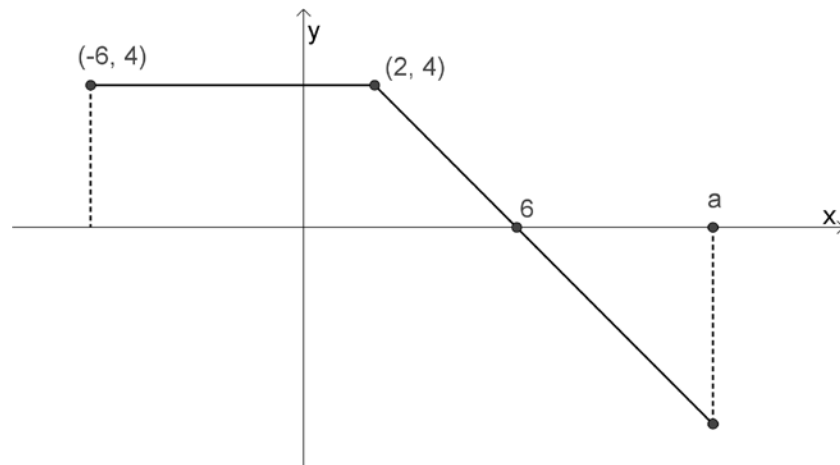
(B) $x < -\frac{1}{2}, x > -\frac{1}{3}$

(C) $x < -3, x > \frac{2}{3}$

(D) $-3 < x < \frac{2}{3}$

Question 8 on the next page

Q8. Using the graph of $y = f(x)$ below,



NOT TO SCALE

determine the value of a which satisfies the condition:

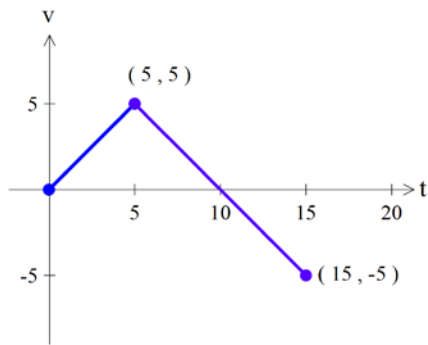
$$\int_{-6}^a f(x) dx = 8$$

- (A) 8
- (B) 10
- (C) 12
- (D) 14

Question 9 on the next page

Q9. A particle is moving along the x -axis.

The graph shows its velocity v metres per second at time t seconds.



When $t = 0$, the displacement x is equal to 5 metres.

What is the maximum value of the displacement x ?

- (A) 12.5 m
- (B) 25.0 m
- (C) 30.0 m
- (D) 47.5 m

Q10. The derivative of $y = x^x$ is

- (A) $x \cdot x^{x-1}$
- (B) $(1 + \log_e x) \cdot x^x$
- (C) $2x \cdot x^x$
- (D) $(x \log_e x) \cdot x^x$

End of Section I

Section II – Short Answer 90 marks

Question 11 (15 marks) Commence on a NEW page.

Marks

(a) Rationalise the denominator $\frac{1 - \sqrt{3}}{5 - \sqrt{3}}$

2

(b) Derive the value $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$

2

(c) Solve $|2x + 1| > 6$

2

(d) Sketch the region defined by the intersection of the inequalities:

3

$$y \geq (x - 1)^3$$

$$y \leq \sqrt{1 - x^2}$$

(e) Differentiate $y = 4x^3 - \sqrt{x}$

2

(f) Find $\int \left(x^3 - \frac{2}{x} \right) dx$

2

(g) Evaluate $\int_0^\pi \sin 2t \, dt$

2

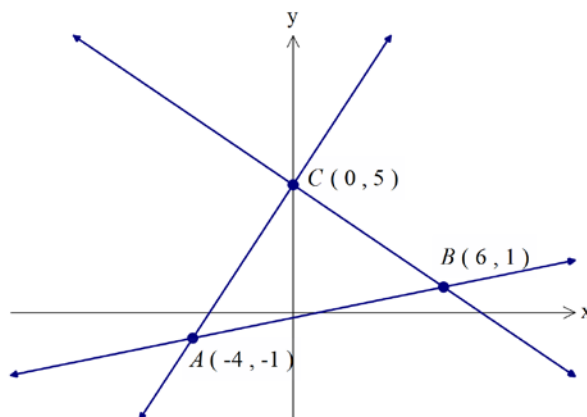
End of Question 11

Question 12 (15 marks)

Commence on a NEW page.

Marks

- (a) The points $A(-4, -1)$, $B(6, 1)$, $C(0, 5)$ are defined in the Cartesian plane.



- | | |
|--|---|
| i) Show that the line passing through points A and B is $x - 5y - 1 = 0$ | 1 |
| ii) Find the distance AB. | 1 |
| iii) Find the area of the triangle $\triangle ABC$. | 3 |
- (b) Differentiate:
- | | |
|---------------------------------------|---|
| i) $y = \cos(7 - x^4)$ | 2 |
| ii) $y = \log_e \frac{2x + 1}{x - 1}$ | 2 |
- (c) Find:
- | | |
|-----------------------------------|---|
| i) $\int 3e^{-5x} dx$ | 2 |
| ii) $\int x^2(1 - \sqrt{x}) dx$ | 2 |
| iii) $\int \frac{6x}{x^2 - 1} dx$ | 2 |

End of Question 12

Question 13 (15 marks) Commence on a NEW page.

Marks

- (a) Sketch the parabola

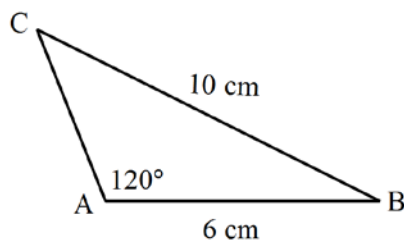
4

$$y^2 + 8x - 2y + 25 = 0$$

clearly showing the location of the vertex, the focus point and the directrix.

- (b) The diagram shows triangle ABC with sides $AB = 6$ cm, $BC = 10$ cm, and $\angle CAB = 120^\circ$.

4



Find the exact value of $\tan C$.

- (c) For what values of k does the line $y = 5x + k$ intersect with the curve $y = x^2 + 3$?

3

- (d) Solve the equation $4 \sin^2 x + 6 \operatorname{cosec}^2 x = 11$, $0 \leq x \leq 2\pi$.

4

End of Question 13

Question 14 (15 marks) Commence on a NEW page.

Marks

(a) Given the function $f(x) = 6x^3 - x^4$

- | | | |
|------|--|---|
| i) | Find the coordinates of the points where the curve crosses the axes | 1 |
| ii) | Find the coordinates of the stationary points and determine their nature | 4 |
| iii) | Find the coordinates of the points of inflexion | 2 |
| iv) | Sketch the graph of $y = f(x)$, clearly indicating the intercepts, stationary points and points of inflexion. | 3 |

(b) In the BBC television documentary “Inside the Factory”, a production manager described the process of manufacturing bulk quantities of baker’s yeast.

He stated that, “the yeast doubles every 3 hrs” and that, “it takes $2\frac{1}{2}$ days to fill the 30,000 kg capacity fermentation vat”.

The growth of the yeast is modelled using the equation,

$$P = P_0 e^{kt}$$

where P is the mass of the yeast in kilograms at time t in hours, and P_0 is the initial amount of yeast put into the fermenter.

- | | | |
|------|---|---|
| i) | Find the exact value of k that produces a doubling of mass every 3 hours. | 2 |
| ii) | What is the mass of the yeast in grams put into the vat at the beginning of the fermentation? | 2 |
| iii) | At what rate is the yeast increasing when there is 12,000 kg of yeast in the tank? | 1 |

End of Question 14

Question 15 (15 marks) Commence on a NEW page.

Marks

(a) A function is defined:

$$f(x) = \begin{cases} \tan x, & 0 < x \leq \frac{\pi}{4} \\ 1, & \frac{\pi}{4} < x < \frac{3\pi}{4} \\ -\tan x, & \frac{3\pi}{4} \leq x \leq \pi \end{cases}$$

i) Sketch the graph of $y = f(x)$. 1

ii) Show that

$$\int \tan x \, dx = -\log_e(\cos x) + C$$
 1

iii) Find the area bounded by $y = f(x)$, the x -axis, $x = 0$ and $x = \pi$. 4iv) The curve $y = f(x)$ is rotated about the x -axis. 4Find the volume of the solid of revolution between $x = 0$ and $x = \pi$.(b) The acceleration of a particle travelling along the x -axis is given by the equation

$$\ddot{x} = 6t - 18$$

where t is the time in seconds, and the acceleration is measured in m/s^2 .The particle has an initial velocity of 15 m/s .i) Find the velocity at time t . 2

iii) At what times does the particle change direction? 1

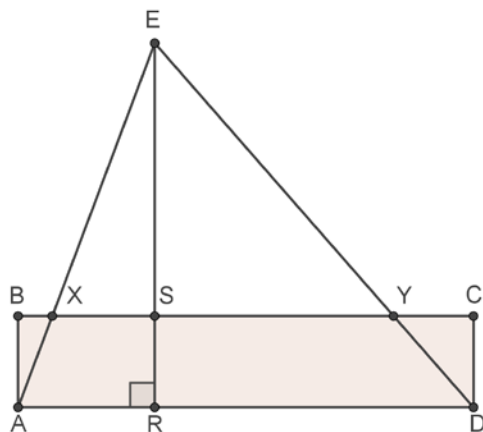
ii) What is the total distance travelled in the first 2 seconds? 2

End of Question 15

Question 16 (15 marks) Commence on a NEW page.

- (a) A triangle AED is constructed using the base of a rectangle ABCD, with intersection points X and Y as shown. ER is the altitude of the triangle AED.

The area of triangle AED is twice the area of the rectangle.

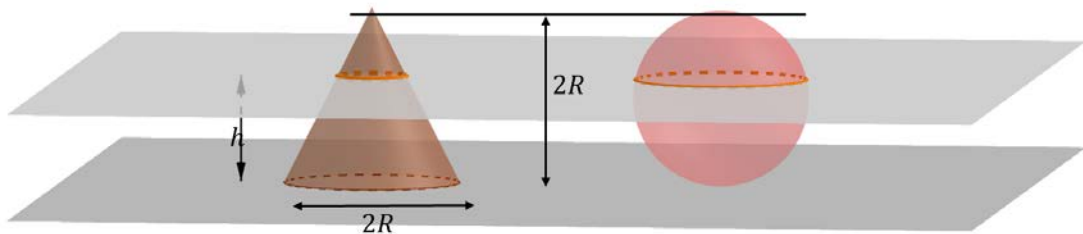


- | | | |
|------|--|---|
| i) | Explain why $ER:ES = 4:3$ | 2 |
| ii) | Prove that $\triangle AED \parallel \triangle XEY$. | 3 |
| iii) | Hence or otherwise, show that $4BX + 4YC = AD$. | 3 |

Question 16 continues on the next page

- b) A sphere of radius R and a right circular cone with radius R at the base and height $2R$ are sitting on a horizontal plane.

A second horizontal plane, height h above the first plane, slices through the sphere and the cone, creating two circular cross sections in the sphere and the cone.



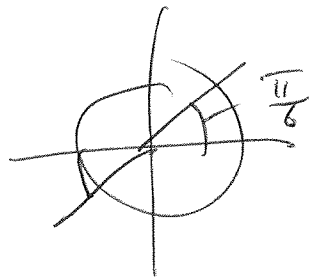
Let the radius of the cross-section in the sphere be r_s and the radius of the cross-section in the cone, r_c .

- i) Show that for cross section of the sphere, $r_s^2 = 2Rh - h^2$ 2
- ii) Show that for the cross section of the cone $r_c^2 = \left(R - \frac{h}{2}\right)^2$ 2
- iii) Hence or otherwise find the height of the slicing plane which gives the maximum sum of cross-sectional areas. 3

End of paper.

Section 1. Multiple Choice,

Q1. $\csc \frac{7\pi}{6} = \frac{1}{\sin \frac{7\pi}{6}}$
 $= -\frac{1}{\frac{1}{2}}$
 $= -2$ (A)



Q2. $\sum_{i=10}^{21} 2i - 30$

$n = (21 - 10) + 1 = 12$ terms
 $\text{first} = 2(10) - 30 = -10$
 $\text{last} = 2(21) - 30 = 12$
 $S_n = \frac{12}{2}(-10 + 12)$
 $= \frac{12}{2} \times 2$
 $= 12$ (D)

Q3. $3x + 4y + 7 = 0$
 perp. line is
 $4x - 3y + C = 0$
 $8x - 6y + 20 = 0$ (C)

Q4. $h = \frac{3-1}{4} = \frac{1}{4}$

Area $\approx \frac{1}{4} (1f_0 + 2f_1 + 2f_2 + 2f_3 + 2f_4 + f_5)$
 $\frac{1}{4} (1 \log_e 1 + 2 \times 1.5 \log_e 1.5 + 2 \times 2 \log_e 2 + 2 \times 2.5 \log_e 2.5 + 1 \times 3 \log_e 3)$
 (B)

- Q1. A
 Q2. D.
 Q3. C.
 Q4. B.
 Q5. B.
 Q6. D.
 Q7. C.
 Q8. D.
 Q9. C
 Q10. B.

Q5. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ in radians.

$$x \text{ radians} = \frac{\pi}{180} x \text{ degrees}$$

$$\lim_{x \rightarrow 0} \frac{\pi}{180} \left(\frac{\sin x}{x} \right) = 0.0174 \quad \text{(B)}$$

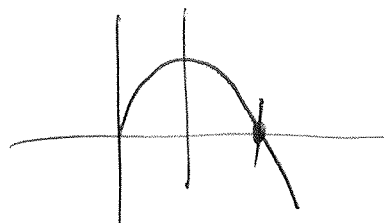
Q6. $\frac{dp}{dt} = 40 \sin(0.1t)$

$$T = \text{Period} = \frac{2\pi}{0.1} = 62.83$$

$$40 \sin 0.1t < 0$$

$$\text{when } t = \frac{T}{2} = 31.41$$

after day 31 ~ so day 32



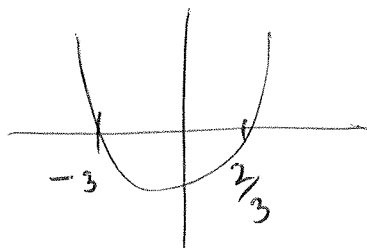
(D)

[Day 1: $0 < t < 1$]

Q7. $3x^2 + 7x - 6 = 0$
 $(3x - 2)(x + 3) = 0$

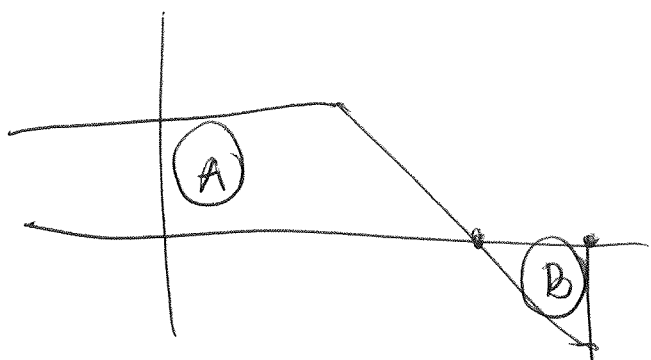
$$x = 2/3 \quad x = -3$$

$$\therefore x < -3 \text{ or } x > 2/3$$



(C)

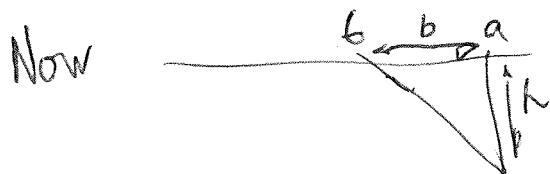
Q8



Area (A) is a trapezium $= \frac{1}{2}(8+12) \times 4 = 10 \times 4 = 40 \text{ u}^2$

Want (A) $\neq$ (B) = 8

$\therefore$ Area B = $\approx 32 \text{ u}^2$



$$\therefore \frac{1}{2} \cdot b \cdot h = 32$$

But gradient = 1, $\therefore b = h$

$$\therefore \frac{1}{2} b^2 = 32$$

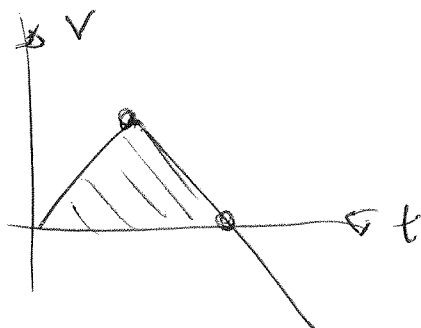
$$b^2 = 64$$

$$b = 8$$

$$a = 6 + 8 = \underline{14}$$

$\therefore$ Answer (D)

Q.9.



Max value x - when $v=0$ (turns around)

$$\begin{aligned} x &= \int v \, dt + C \\ &= \text{area triangle} + C \\ &= \frac{1}{2}(10)(5) + C \\ &= 25 + C \end{aligned}$$

Now $x(0) = 5$

$$\begin{aligned} \therefore x_{\max} &= 25 + 5 \\ &= 30 \text{ m.} \end{aligned} \quad \text{(C)}$$

Q10.

$$\begin{aligned} \rightarrow y &= x^n \\ &= (e^{\ln x})^n \\ &= e^{(n \ln x)} \end{aligned}$$

$$\begin{aligned} y' &= e^{(n \ln x)} \cdot \frac{d}{dx}(n \ln x) \\ &= e^{n \ln x} \cdot (1 \times \ln x + x \times \frac{1}{x}) \\ &= (1 + \ln x) e^{n \ln x} \\ &= (1 + \ln x) x^n \end{aligned} \quad \text{(B)}$$

Section II.

$$\text{Q11. (a) } \frac{1-\sqrt{3}}{5-\sqrt{3}} = \frac{1-\sqrt{3}}{5-\sqrt{3}} \times \frac{5+\sqrt{3}}{5+\sqrt{3}}$$

$$= \frac{(1-\sqrt{3})(5+\sqrt{3})}{25-3}$$

✓ ✓

$$= \frac{5+5\sqrt{3}-5\sqrt{3}-3}{22}$$

$$= \frac{2-4\sqrt{3}}{22}$$

✓ ✓

$$= \frac{1-2\sqrt{3}}{11}$$

$$(b) \lim_{x \rightarrow 3} \frac{x^3-27}{x^2-9} = \lim_{x \rightarrow 3} \frac{(x-3)(x^2+3x+9)}{(x-3)(x+3)}$$

✓ ✓

$$= \lim_{x \rightarrow 3} \frac{x^2+3x+9}{x+3}$$

$$= \frac{(3)^2+3(3)+9}{3+3}$$

$$= \frac{27}{6}$$

$$= \frac{9}{2}$$

✓ ✓

MUST show working
for 2 marks.
question said "derive"

Q11.

(c) $|2x+1| > 6$

$$2x+1 > 6 \quad \text{or} \quad -(2x+1) > 6$$

$$2x+1 < -6$$

$$2x > 5$$

$$2x < -7$$

$$x > 5/2 \quad \text{or}$$

$$x < -7/2$$

✓ 1 per solution ✓

(d) $y \geq (x-1)^3$

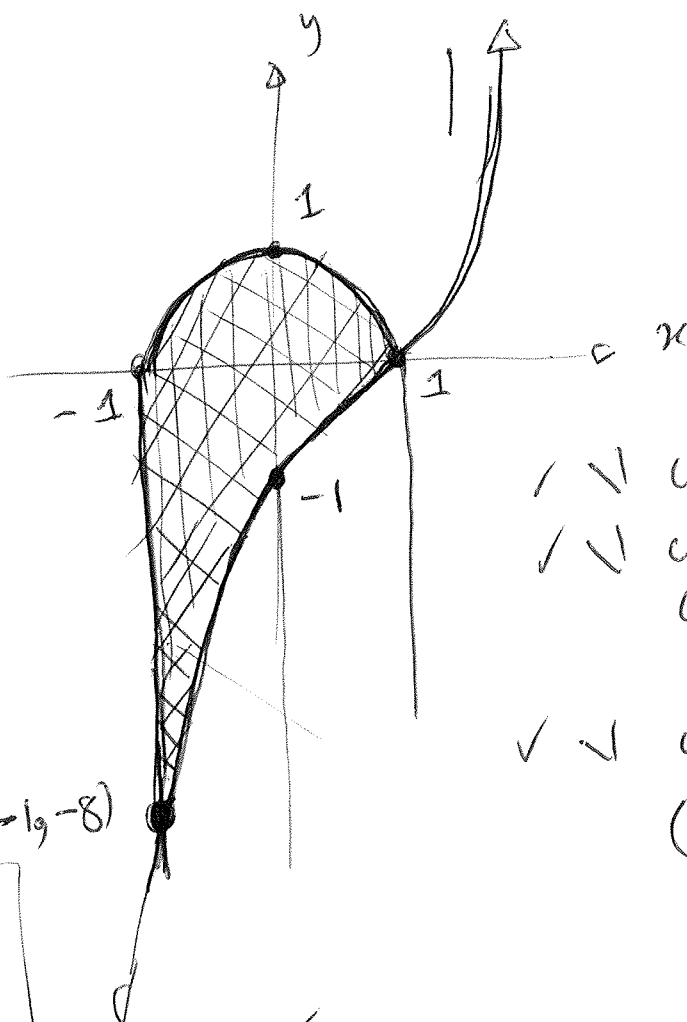
$$y \leq \sqrt{1-x^2}$$

(e) $y = 4x^3 - \sqrt{x}$
 $= 4x^3 - x^{1/2}$

$$y' = 12x^2 - \frac{1}{2}x^{-1/2}$$

$$= 12x^2 - \frac{1}{2\sqrt{x}}$$

✓ ✓
 12 marks - deduct 1 per error.



✓ ✓ correct semi circle
 ✓ ✓ correct cubic (with intercept on y)
 ✓ ✓ correct region
 (-1, -8) not required.

(f) $\int (x^3 - \frac{2}{x}) dx = \frac{1}{3}x^3 - 2 \log_e x + C$

12 marks
 Deduct one per error.
 Must have $+C$ for full marks

(g) $\int_0^\pi \sin 2t dt = \left[-\frac{1}{2} \cos 2t \right]_0^\pi$ ✓ ✓
 $= -\frac{1}{2} (\cos 2\pi - \cos 0)$
 $= 0$ ✓ ✓

Q12. $A(-4, -1), B(6, 1), C(0, 5)$

(i) $x - 5y - 1 = 0$

Point A: $(-4) - 5(-1) - 1 = -4 + 5 - 1 = 0 \quad \therefore A \text{ on the line.}$

B: $(6) - 5(1) - 1 = 6 - 5 - 1 = 0 \quad \therefore B \text{ on the line}$ ✓ ✓

(ii) Distance AB:

$$d^2 = (6 + 4)^2 + (2)^2$$

$$= 100 + 4$$

$$= 104$$

$AB = \sqrt{104} \text{ units.}$ ✓ ✓

$= 2\sqrt{26} \quad \leftarrow (\text{optimal})$

(iii) Perp height AB to C B $AB: x - 5y - 1 = 0$
 $C: (0, 5)$

$$d = \frac{|(0) - 5(5) - 1|}{1^2 + 5^2}$$

$$= \frac{26}{\sqrt{26}}$$

✓ using perp. distance.

✓ correct value.

$$\therefore \text{Area } \triangle ABC = \frac{1}{2} (2\sqrt{26}) \times \frac{26}{\sqrt{26}}$$

$$= \underline{26 \text{ units}^2}$$

✓ using (ii) with perp.

Q12 (b)

(i) $y = \cos(7-x^4)$

$$y' = -\sin(7-x^4) \cdot (-4x^3)$$

$$= 4x^3 \sin(7-x^4)$$

✓✓

(ii) $y = \log\left(\frac{2x+1}{x-1}\right)$

$$= \log(2x+1) - \log(x-1)$$

✓ Easy way

$$y' = \frac{1}{2x+1} - \frac{1}{x-1}$$

✓

OR

$$y' = \frac{(x-1)}{(2x+1)} \times \frac{2(x-1) - (2x+1)}{(x-1)^2}$$

Hard way - chain rule, quotient rule.

$$= \frac{(x-1)}{(2x+1)} \times \frac{-3}{(x-1)^2}$$

$$= \frac{-3}{(2x+1)(x-1)}$$

(c) (i) $\int 3e^{-5x} dx = -\frac{3}{5}e^{-5x} + C$

(ii) $\int x^2(1-\sqrt{x}) dx = \int (x^2 - x^2\sqrt{x}) dx$

✓ for expanding

$$= \int (x^2 - x^{5/2}) dx$$

$$= \frac{1}{3}x^3 - \frac{2}{7}x^{7/2} + C$$

✓ correct answer.

(iii) $\int \frac{6x}{x^2-1} dx = 3 \log_e(x^2-1) + C$

✓✓

Q13

(a) $y^2 + 8x - 2y + 25 = 0$

$$y^2 - 2y = -8x - 25$$

$$(y-1)^2 - 1 = -8x - 25$$

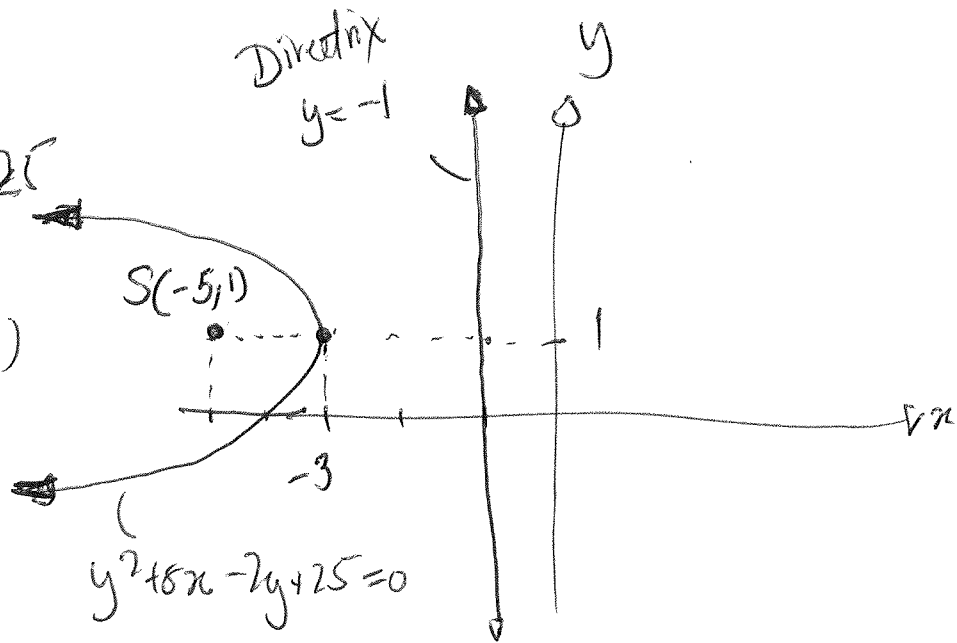
$$(y-1)^2 = -8x - 24$$

$$(y-1)^2 = -8(x+3)$$

Vertex $(-3, 1)$

$$a = 2$$

✓ complete
squares
correctly



- ✓ correct orientation
- ✓ correct vertex
- ✓ correct focus and directrix

(b) $\frac{\sin 120^\circ}{10} = \frac{\sin C}{6}$

$$\sin C = \frac{6 \times \sin 120^\circ}{10}$$

$$= \frac{3}{5} \times \frac{\sqrt{3}}{2}$$

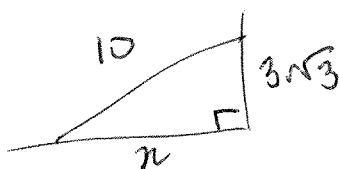
$$= \frac{3\sqrt{3}}{10}$$

✓ correct use sine rule

✓ correct value $\sin C$

✓ find hypotenuse

✓ correct value $\tan C$



$$x^2 = 100 - 27 = 73$$

$$\therefore \tan C = \frac{3\sqrt{3}}{\sqrt{73}}$$

$$(1) \quad y = 5x + k \quad \text{and} \quad y = x^2 + 3$$

Intersect where

$$5x + k = x^2 + 3$$

$$x^2 - 5x + (3 - k) = 0$$

✓ Form equation

Require $\Delta \geq 0$

$$\Delta = b^2 - 4ac$$

$$= 25 - 4(1)(3 - k)$$

$$= 25 - 12 + 4k$$

$$= 13 + 4k$$

✓ Use discriminant correctly

$$\therefore 13 + 4k \geq 0$$

$$k \geq -\frac{13}{4}$$

✓ Find value.

$$(2) \quad 4 \sin^2 x + 6 \csc^2 x = 11$$

$$4 \sin^2 x + \frac{6}{\sin^2 x} = 11$$

Let $u = \sin^2 x$

$$4u + \frac{6}{u} = 11$$

$$4u^2 + 6 = 11u$$

✓ $\rightarrow 4u^2 - 11u + 6 = 0$

$$4u^2 - 8u - 3u + 6 = 0$$

$$4u(u - 2) - 3(u - 2) = 0$$

$$(4u - 3)(u - 2) = 0$$

$$0 \leq x \leq 2\pi$$

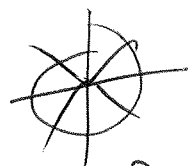
$$\therefore 4 \sin^2 x - 3 = 0 \quad \text{or} \quad \csc^2 x - 2 = 0$$

$$\therefore \sin^2 x = \frac{3}{4}$$

$$\sin x = \pm \frac{\sqrt{3}}{2}$$

Not possible

✓ for rejection



$$x = \frac{7\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

✓✓ I made per pair of solutions - (some will forget

$$(\pm \sqrt{3}/2)$$

Q14-

(a) $f(x) = 6x^3 - x^4$

(i) $6x^3 - x^4 = 0$

$$x^3(6-x) = 0$$

x intercepts at $(0,0), (6,0)$ ✓

(ii) $f'(x) = 18x^2 - 4x^3$
 $= 2x^2(9x - 2x)$

$\therefore f'(x) = 0$ at $x=0, x=9/2$
 $(0,0) \quad (9/2, \frac{2187}{16})$

✓✓ find two stationary points

$$f''(x) = 36x - 12x^2$$

x	0	9/2
$f'(x)$	0	0
$f''(x)$	0	-81

Can't tell.

local max at $(9/2, \frac{2187}{16})$

✓✓ determine nature.

x	-1	0	1
$f'(x)$	22 /	0 —	14 /

horizontal point of inflexion at $(0,0)$

$$(iii) f''(x) = 36x - 2x^2$$

$$= 12x(3 - x)$$

Possible inflexion at $x=0, x=3$

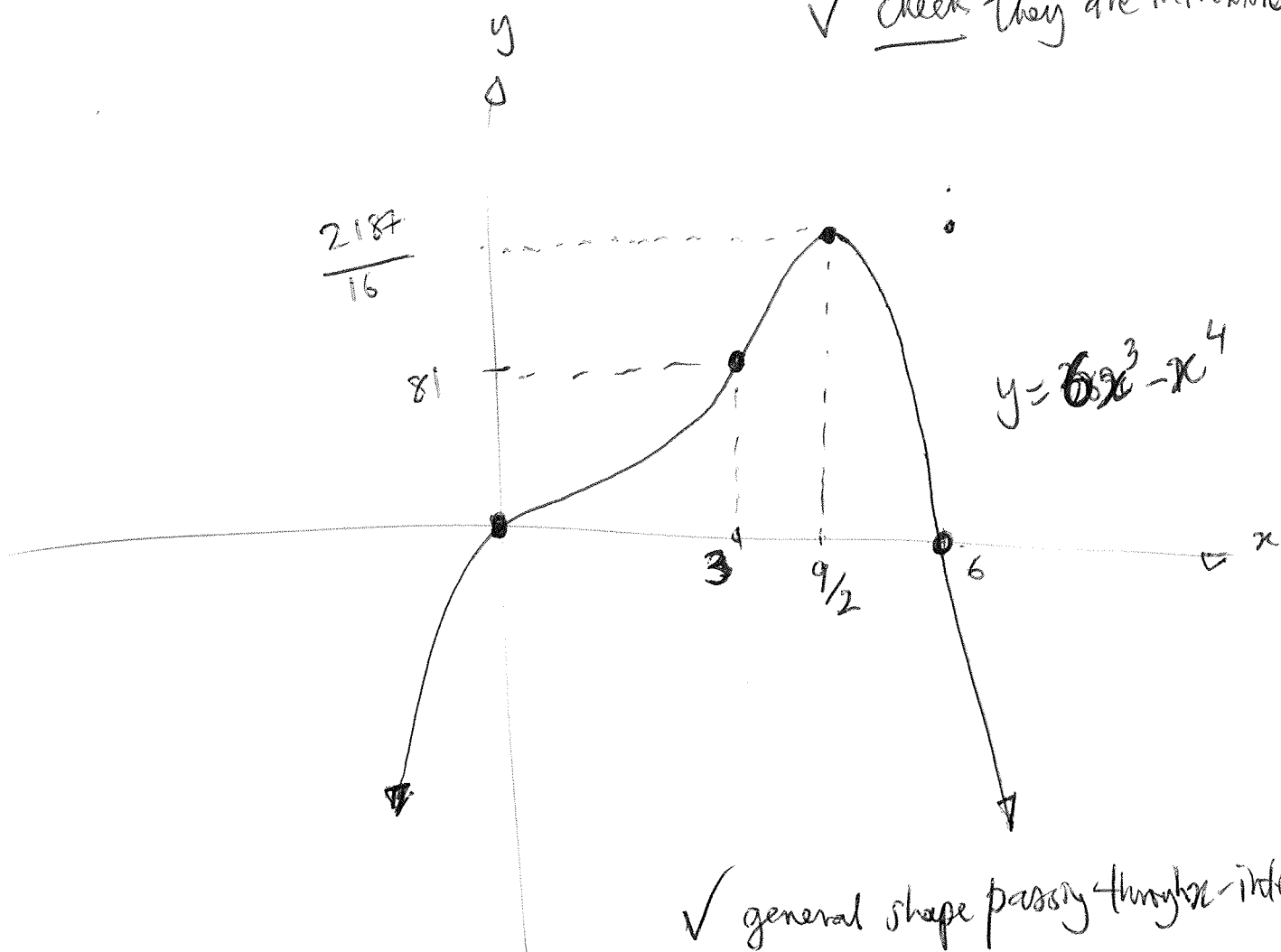
already shown inflexion

x	1	3	4
$f''(x)$	24	0	-48
	U		∩

changes concavity

∴ Inflexion points at $(0,0), (3,81)$

✓ find points of inflexion
✓ check they are inflexions!



✓ general shape passing through intercepts

✓ correct local max shown

✓ inflexions clearly showing
at $(0,0), (9/2, \frac{2187}{16})$

Q14(b)

$$P = P_0 e^{kt}$$

(i) Double in 3 hours:

$$2P_0 = P_0 e^{3k}$$

$$2 = e^{3k}$$

$$\ln 2 = 3k$$

$$k = \frac{\ln 2}{3}$$

$$(ii) 30,000 = P_0 e^{(\frac{\ln 2}{3})(60)}$$

$$P_0 = \frac{30,000}{e^{20 \ln 2}}$$

$$\approx 0.0286 \text{ kg}$$

$$\approx 28.6 \text{ gms.}$$

$$2.5 \text{ days} = 60 \text{ hrs}$$

✓ correct substitution

✓ correct answer in grams.

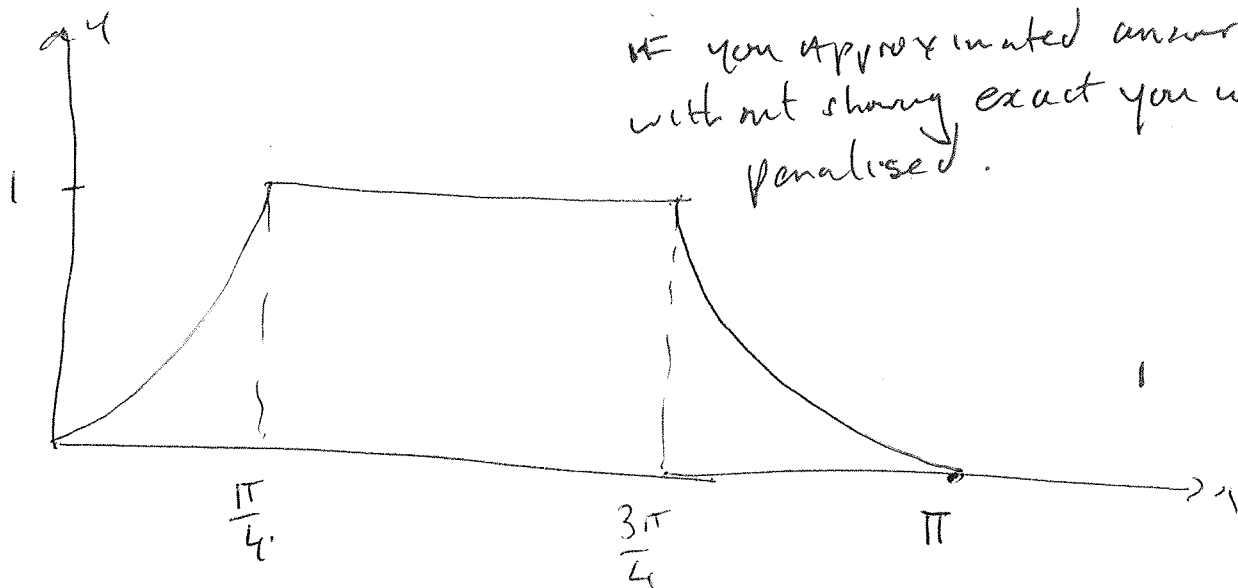
$$(iii) \frac{dP}{dt} = k \cdot P$$

$$= \frac{\ln 2}{3} \cdot 12,000$$

$$\approx 2773 \text{ kg/hr}$$

✓

15a) 11

1
2
3
4
5
6
7
8

If you approximated answer without showing exact you were penalised.

$$ii) \int_{\pi/4}^{\pi} \tan x \, dx = \int_{\pi/4}^{\pi} \frac{\sin x}{\cos x} \, dx = -\ln(\cos x) + c$$

$$iii) A = 2 \int_{\pi/4}^{\pi} \tan x \, dx + \frac{\pi}{2} \times 1$$

$$= 2 \left[-\ln(\cos x) \right]_{\pi/4}^{\pi} + \frac{\pi}{2}$$

$$= 2 \left[-\ln \frac{1}{2} - (-\ln 1) \right] + \frac{\pi}{2}$$

$$= 2 \ln \sqrt{2} + \frac{\pi}{2} = \ln 2 + \frac{\pi}{2}$$

$$iv) v = \pi \int_0^{\pi} y^2 \, dx = 2\pi \int_{\pi/4}^{\pi} \tan^2 x \, dx + \pi \int_0^{\pi/4} 1 \, dx + \pi \int_{3\pi/4}^{\pi} \tan^2 x \, dx$$

$$= 2\pi \int_{\pi/4}^{\pi} (\sec^2 x - 1) \, dx + \pi \left[x \right]_0^{\pi/4} + \pi \int_{3\pi/4}^{\pi} (\sec^2 x - 1) \, dx$$

$$= 2\pi \left[\tan x - x \right]_{\pi/4}^{\pi} + \pi \left[\frac{3\pi}{4} - \frac{\pi}{4} \right] + \pi \left[\tan x - x \right]_{3\pi/4}^{\pi}$$

$$= \pi \left[1 - \frac{\pi}{4} \right] + \frac{\pi^2}{2} + \pi \left[(0 - \pi) - (-1 - \frac{3\pi}{4}) \right]$$

$$= \pi - \frac{\pi^2}{4} + \frac{\pi^2}{2} + \pi \left[-\pi + 1 + \frac{3\pi}{4} \right]$$

$$= \pi - \frac{\pi^2}{4} + \frac{\pi^2}{2} + \pi - \frac{\pi^2}{4}$$

$$= \underline{\underline{2\pi}}$$

b) $\dot{x} = 6t - 18$ Q15(b)

i) $\dot{x} = 3t^2 - 18t + c$

$t=0 \quad \dot{x} = 15$

$c = 15$

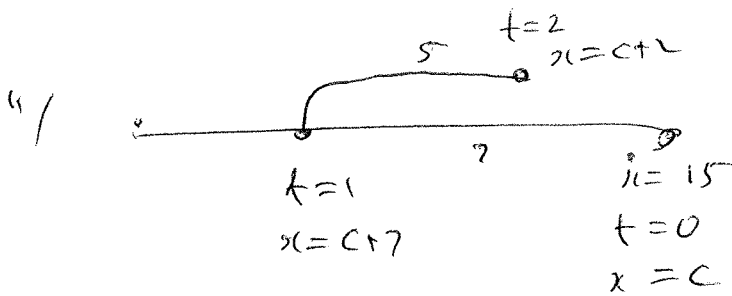
$\dot{x} = 3t^2 - 18t + 15$ ✓

(1, if no 15)

iii) $\dot{x} = 0 \quad t^2 - 6t + 5 = 0$

$(t-1)(t-5) = 0$

$t=1$ or $t=5$ changes direction at $t=1$ + $t=5$



1

$x = t^3 - 9t^2 + 15t + C$

$t=0 \quad x=C$

$t=1 \quad x=7+C$

$t=2 \quad x = 8 - 36 + 30 + C$
 $= 2 + C$
 $= C+2$

So Distance travelled = $7+5$

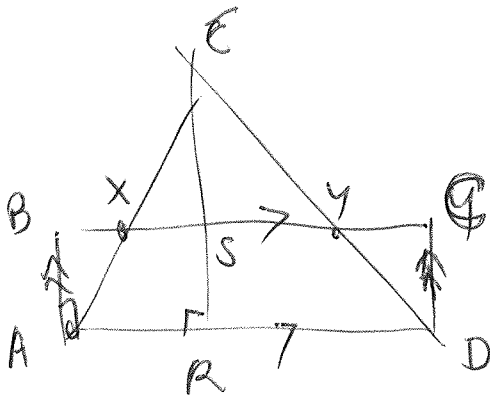
= ~~8~~ 12

~~8~~ 1

integrate but
don't break up
Area 1

Q 16.

(a)



$$\text{Area } \triangle AED = 2 \times \text{Area } ABCD$$

$$\frac{1}{2} (AD) (ER) = 2 \times AD \times SR$$

$$AD \cdot ER = 4SR$$

$$ER = 4(ER - ES)$$

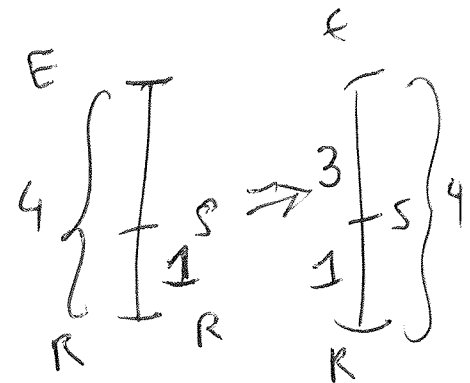
$$ER = 4ER - 4ES$$

$$4ES = 3ER$$

$$\frac{4}{3} = \frac{ER}{ES}$$

$$\therefore ER:ES = 4:3$$

✓



✓

sufficient to
show $ER = 4SR$
then conclude
 $\frac{ER}{ES} = \frac{4}{3}$

(b) In $\triangle AED$ & $\triangle XEY$,

$$\angle AED = \angle XEY \text{ (common)}$$

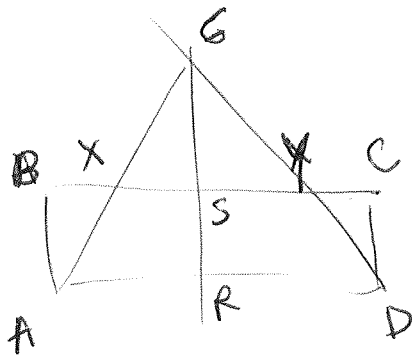
~~$\angle AED$~~

$$\angle EAD = \angle EXY \text{ (} BC \parallel AD \text{ - opp sides rectangle } \checkmark$$

$$\therefore \triangle AED \parallel \triangle XEY \text{ (equiangular)}$$

✓

Q16(a)
(ii)



Since $\triangle AED \parallel \triangle XEY$,
matching sides are in proportion,

$$\therefore \frac{XY}{AD} = \frac{3}{4} \quad \left(\text{from } \frac{ES}{ER} = \frac{3}{4} \right)$$

✓ using
sim.
triangle
ratio

$$\therefore \frac{AD - BX - CY}{AD} = \frac{3}{4} \quad (BC = AD \text{ opposite sides rectangle.})$$

$$4(AD - BX - CY) = 3AD$$

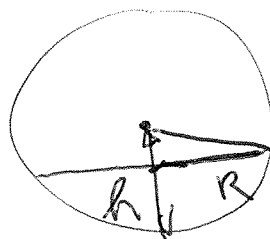
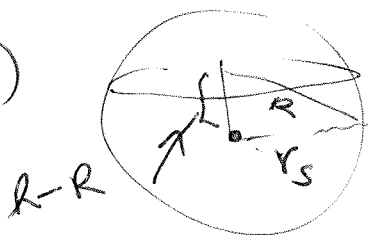
$$4AD - 3AD = 4BX + 4CY$$

$$AD = 4BX + 4CY \text{ as required}$$

✓✓ for working
for conclusion

Q16(b)

(i)



Using Pythagoras

$$R^2 = (R-h)^2 + r_s^2$$

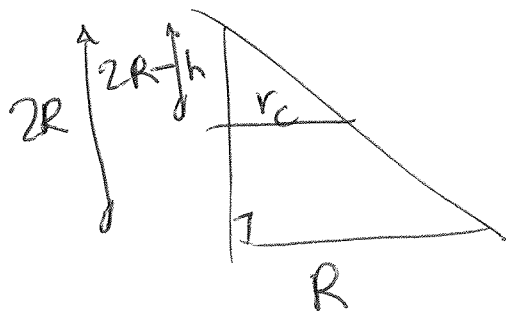
$$r_s^2 = R^2 - (R^2 - 2Rh + h^2)$$

$$r_s^2 = 2Rh - h^2 \text{ as required}$$

✓

✓

Q16. (b) (ii)



similar triangles (equiangular)

$$\therefore \frac{r_c}{R} = \frac{2R-h}{2R}$$

✓

$$r_c = R - \frac{h}{2}$$

$$r_c^2 = \left(R - \frac{h}{2}\right)^2$$

✓

$$(iii) \text{ Area total} = \pi(r_s^2 + r_c^2)$$

$$= \pi(2Rh - h^2 + \left(R - \frac{h}{2}\right)^2)$$

$$= \pi(2Rh - h^2 + R^2 - Rh + \frac{h^2}{4})$$

$$= \pi\left(R^2 - \frac{3h^2}{4} + Rh\right)$$

✓

$\frac{dA}{dh}$

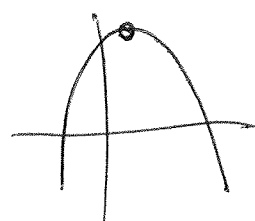
$\approx \Delta A$

$\approx \Delta A$

quadratic in h , $h < 0$

$$\text{Maximum at } x = \frac{-b}{2a} \Rightarrow h = \frac{-R}{2(-3/4)}$$

$$\boxed{h = \frac{2}{3}R}$$



✓ - justify maximum by graph or first derivative

✓

$$\text{or: } \frac{dA}{dh} = \pi\left(-\frac{3}{2}h + R\right)$$

$$\frac{dA}{dh} = 0 \text{ when } h = \frac{4}{6}R = \frac{2}{3}R$$

$$\frac{d^2A}{dh^2} = -\frac{3\pi}{2} < 0 \text{ always } \therefore \text{maximum value.}$$